Dietary Epidemiology of Essential Tremor: Meat Consumption and Meat Cooking Practices

Background/Aim Harmane [1-methyl-9H-pyrido(3,4-b)indole] is a tremor-producing neurotoxin. Blood harmane concentrations are elevated in essential tremor (ET) patients for unclear reasons. Potential mechanisms include increased dietary harmane intake (especially through well-cooked meat) or genetic-metabolic factors. We tested the hypothesis that meat consumption and level of meat doneness are higher in ET cases than in controls. Methods Detailed data were collected using the Lawrence Livermore National Laboratory Meat Questionnaire. Results Total current meat consumption was greater in men with than without ET (135.3 ± 71.1 vs. 110.6 ± 80.4 g/day, $p = 0.03$) but not in women with versus without ET (80.6 ± 50.0 vs. 79.3 ± 51.0 g/day, $p = 0.76$). In an adjusted logistic regression analysis in males, higher total current meat consumption was associated with ET (OR = 1.006, $p = 0.04$, i.e., with 10 additional g/day of meat, odds of ET increased by 6%). Male cases had higher odds of being in the highest than lowest quartile of total current meat consumption (adjusted OR = 21.36, $p = 0.001$). Meat doneness level was similar in cases and controls. Conclusion This study provides evidence of a dietary difference between male ET cases and male controls. The etiological ramifications of these results warrant additional investigation.